

NOTICEDSK GST DRAFTING ENGINE

3-Tier Cascade Architecture & Automated Testing Strategy Report

1. Executive Summary

NoticeDesk is an enterprise-grade litigation operating system engineered for Indian Chartered Accountancy and legal practices. The core drafting engine automates the generation of filing-grade replies to GST Show Cause Notices (SCNs), audit objections, and assessment orders. To optimize cost, execution latency, and legal precision, NoticeDesk employs a 3-Tier Cascade Architecture that intelligently routes incoming notices based on similarity matching against a 301-pair GST corpus, vector RAG retrieval, and Claude LLM generation.

2. 3-Tier Cascade Routing Architecture

The classification engine (**GstCorpusMatcherService**) evaluates incoming GST notice OCR and parsed metadata against NoticeDesk's 301 paired corpus records (NoticeDesk GST Corpus 1-300 Paired) using three distinct execution paths:

Tier 1: Fast-Track Direct Match (Similarity ≥ 90%)

When an incoming notice matches an existing corpus notice with ≥ 90% confidence, the system bypasses expensive LLM calls. The Fast-Track Engine (**GstTemplateFillerService**) extracts the pre-verified corpus .docx template and programmatically replaces placeholders (GSTIN, PAN, Legal Name, Notice Number, DIN, Authority, Issue Date, State) across all 15 standard legal sections using Apache POI. **Result: Zero LLM token cost and sub-second (<1s) latency.**

Tier 2: Novel Notice Handling (RAG Score < 70% & Post-Draft Auto-Caching)

When a notice presents rare or unprecedented legal issues with no matching precedent (< 70% similarity), the system routes the notice directly to Claude LLM via **DraftingAgent** and the **drafting_v4** system prompt. After generating a fresh 15-section draft, the system automatically caches the generated output back into the RAG vector store (**RagStoreService.cacheNovelDraft**) under legal_chunks. This ensures future similar notices hit Tier 1 or Tier 3.

Tier 3: Hybrid Multi-Issue Notice Handling (Similarity 70% - 89%)

For hybrid notices combining multiple legal issues (e.g., Section 129 detention combined with E-way bill breakdown), the system seeds all 301 corpus pairs into the RAG vector store via **CorpusRagSeederService**. It retrieves matching legal chunks and passes them to Claude solely for formatting, legal synthesis, and section alignment, avoiding unguided generation from scratch.

3. Standard 15-Section Legal Framework

All output drafts generated across all three tiers strictly adhere to the professional 15-section GST litigation structure defined in **drafting_v4.md**:

#	Section Title	Legal Purpose & Contents
1	Addressee & Reference Block	To / designation / jurisdiction / GSTIN / PAN / notice reference & date
2	Subject	One-sentence concise summary of reply and relief sought
3	Synopsis	Executive summary of allegations, legal defenses, and core relief

4	Statement of Facts	Numbered chronological matrix of facts — strictly unargued
5	Preliminary Objections — Jurisdiction	Officer authority, territorial limits, and monetary threshold defects
6	Preliminary Objections — Limitation & Procedure	Section 73(10)/74(10) time bar, Rule 142(1A) DRC-01A lack, DIN invalidation
7	Para-wise Reply	Numbered paragraph-by-paragraph admission / denial / rebuttal
8	Ground 1	Primary substantive ground (e.g. absence of suppression under Pushpam)
9	Ground 2	Secondary substantive ground (e.g. non-denial of ITC for supplier default under Suncraft)
10	Ground 3	Tertiary ground (e.g. format notice natural justice violation under NKAS Services)
11	Without Prejudice — Alternative Submissions	Section 73(8) penalty waiver & Section 50 net cash interest defense
12	Quantum, Interest & Computation	Arithmetic breakdown, rate verification, cash liability restriction
13	Prayer	Numbered specific relief + explicit request for Personal Hearing under Sec 75(4)
14	Annexures	List of attached evidence (Form REG-06, GSTR-1, GSTR-3B, GSTR-2A/2B reconciliations)
15	Declaration & Signature Block	Formal verification statement, authorised signatory details, place, and date

4. Backend Implementation & Validation

The solution is implemented in Spring Boot (**noticedesk-java**) and verified via JUnit test suite

ScenariosTestingRunnerTest. Output files for all three scenarios are published in PDF, DOCX, and TXT formats inside the **/output** directory.